



Zoology Legends

BT101

Current Final TERM Spring 2023

Muhammad Zaman 🖐️ 😊

0304-4756496

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Zoology Legends

MCQS Spring 2023

1. The external factors, which control the ecosystem, are **Independent** of internal factors.
2. Fungi is Primary **decomposer**.
3. Winter Sleep occurs in large animals to **Energy** ... reserves.
4. Y-axis is a logarithmic plot of numbers of survivors X-axis is a linear plot of **age** .
5. The final community is the **Climax** community.
6. Camouflage use of any combination of materials **coloration** or Illumination to avoid predator
7. 541 million years ago, O₂ made **15%** stable of atmosphere.
8. The total average height of troposphere is **13Km** .
9. Adult human body contain **60%** of water.
10. Cascade Effect Removal of **Apex predator** ...in environment.
11. Reptiles are strongly affected by **light** pollution.
12. Bio magnification is process of increasing **concentration** of toxic substance.
13. First evolutionary scheme was proposed by Jean Baptiste Lamarck in **1809** .
14. Extinct hippopotamus like **Taxodon** .



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15. Adaptation are characteristics that **increase the potential** of an organism or species to successfully reproduce in a specified environment.
16. Natural selection ---- **Removes** ----- variations from a population by favoring one allele over other at a gene locus.
17. Which of the following protein is present in the cellular respiration pathways in all eukaryotic organisms. **Cytochrome C.**
18. Short protein appendages smaller than flagella is called **Pilli**
19. Genomic DNA sequence of an Archaeon and --- **Bacterium**- -- were first compared in 1996.
20. Toxoplasmosis caused by toxoplasma virulent in immune suppressed individuals as in **AIDS** .
21. Natural selection is the most powerful of the principal agents of **genetic** change.
22. Term ribosomes was coined by **Palade (1955)** .
23. Natural selection works by **favoring individuals traits** over individuals with alternative traits.
24. Rise in mean temperature of **0.6-1.0°C** in arid coastal areas, arid mountains and hyper arid plains recently in **Pakistan**
25. Unique excretory system of nematoda comprised of two or more **renette** cell.
26. Skeleton of Poriferan consist of microscopic needlelike spikes called **spicules**
27. Lizard belongs to order **squamata**.
28. **Cyanobacteria** have Symbiotic relationship with leguminous plants roots.



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29. Related animals have similar **DNA** derived from their common ancestor.

30. Sweat glands in mammals are called **Sudoriferous** glands.

31. In echinoderms **Ammonia** is the primary nitrogenous waste product.

32. **Porins** allow molecules to freely diffuse from one side of the membrane to the other.

33. Deforestation Directly affects rate of **transpiration**

34. Disruption Variety of causes, independently or in concert, are responsible for **extinctions**

35. Animal that changes its color rapidly . **Chameleon**

36. The resemblance of one species to another for protective or aggressive purposes is **mimicry**

37. **Exotic** Species are species that occur outside of their natural ranges because of human activity.

The final community is the **Climax** community.

First evolutionary scheme was proposed by Jean Baptiste Lamarck in **1809**

Short protein appendages smaller than flagella are called **Pilli** .

A species is a **mosaic** of different molecules and structures that have evolved at different rates

One species gains benefit other species is harmed in **parasitism**.

The atmospheric concentration of CO₂ measured from Antarctic ice core data implies that Earth's climate has been pretty stable over the past **400,000** years.



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Secondary Treatment is a **Biological process**.

Natural selection prefers **large population**.

Only change in **DNA** are transferred to next generations.

Intraspecific competition between members of same species for same resources.

Land pollution indirectly affects the **respiratory system** of human beings.

nitrogen fixing microbes need enzyme **nitrogenase**

Extinct hippopotamus like animals now called __ **Taxodone**

Animal changes its color rapidly ____ **Chameleon**

Above thermosphere is ____ **Exosphere**.

__ is located above the thermosphere. ____ **Exosphere**

----- Rapidly changes its color. __ **Chameleon**

Most reptiles excrete ____ **Uric Acid**

RER has ____ **Ribosomes Attached**.

All living organism 16% nitrogen by weight atmosphere is
nitrogen by volume __ **78%**

Species are species that occur outside of their natural ranges because of human activity ____ **Exotic**

Hydrocarbons Diverse group of organic compounds that contain only and carbon ____ **Hydrogen**.

Industrial may contain acids, alkalis, salts, poisons, oils and in some cases harmful bacteria. ____ **Effluents**



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A species is a of different molecules and structures that have evolved at different rates__ **Mosaic**.

Microbes_ example of chemoautotrophs

the total average height of the troposphere_**13Km**

Toxoplasmosis caused by Toxoplasma virulent in immune suppressed individually_as in__ **AIDS**

animals changes its colours rapidly_**Chameleon**_

Bird voice box_**Syrinx**_

All of the frogs in a particular pond or all of the sea urchins in a particular tidepool make up a_**Deme**

All living organism 16% nitrogen by weight atmosphere is **78%**.... nitrogen by volume.

Industrial **effluents**.... may contain acids, alkalis, salts, poisons, oils and in some cases harmful bacteria.

The cnidocyte has a modified Cilium.... called a **cnidocil**.

Y-axis is a logarithmic plot of numbers of survivors X-axis is a linear plot of __**age**

Nitrogen in -----% in atmosphere.**78%**

Extinct hippopotamus like...**taxodon**

skeleton of Echinodermata consists of a series of calcium carbonate plates called____**Ossicles**

The an organism occupies is the total of all the ways it uses the resources of its environment. _ **Niche**

A species is a of different molecules and structures that have evolved at different rates__ **Mosaic**



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541 million years ago O₂ made stable of atmosphere__ **15%**

Genomic DNA sequence of an Archaean and ----- were first compared in 1996__ **Bacterium**

Natural selection ----- variations from a population by favoring one allele over other at a gene locus. __ **Removes**

....allow molecules to freely diffuse from one side of the membrane to the other. __ **Porins**

----- is removal of forest or stand of trees from land which is then converted to a non-forest use. _ **Deforestation**

Evolutionary changes are occurring ____over millions of years. ____ **Gradually**

The external factors which control the ecosystem are ----- of internal factors _ **Independent**

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Unique excretory system of nematoda comprised of two or more **renette cell**.

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Bird voice box __ **Syrinx**



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Near the surface ozone concentrations about ____ **0.04-0.15ppm**

-----was first isolated and named in 1817 ____ **Chloroplast.**

Activated sludge-sewage mixture is called mixed ____ **Liquor**

The organism occupies is the total of all the ways it uses the resources of its environment. ____ **Niche**

Camillo Golgi country is ____ **Italian**

All living organism 16% nitrogen by weight atmosphere is
nitrogen by volume ____ **78 %**

Natural selection favor ____ **large gene**

Which animal hibernate ____ **bear**

ZOOLOGY
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Repeated Questions Spring 2023

Stages of Water cycle

1. Evaporation
2. Transpiration
3. Condensation
4. Precipitation
5. Surface Run off
6. Infiltration

Characteristics of Bacteria

- ⇒ Average bacteria 0.5 - 2.0 μm in diameter.
- ⇒ Surface Area $\sim 12 \mu\text{m}^2$
- ⇒ Volume is $\sim 4 \mu\text{m}$
- ⇒ Surface Area to Volume is 3:1
- ⇒ Gel-like matrix of water, enzymes, nutrients, wastes and gases and contains cell structures like numerous ribosomes and polysomes.
- ⇒ No endoplasmic reticulum and no membrane bound organelles.
- ⇒ Location of growth, metabolism, and replication.



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Evidence of theory of natural selection

Darwin based his theory on many solid evidences from his observations and collected samples. Three major evidences:

- 1) Geology
- 2) Fossil Evidences
- 3) Galápagos islands

Evidence of the theory of natural selection refers to the observable and empirical data that supports the idea that individual of advantageous traits are more likely to survive, reproduce, and pass on their traits to future generations, resulting in evolutionary changes in a population over time.

This evidence includes observations of variations within populations, adaptation to specific environments, the fossil record showcasing transitional forms, comparative anatomy and embryology, as well as genetic studies and experimental demonstrations.

Structure and Function of Golgi apparatus.

Structure

- Located near the end of the ER close to the nucleus
- Composed of several layers of cisternae (fluid-filled sacs)

Has three different parts:

1. cis-Golgi
2. medial-Golgi
3. trans-Golgi

Functions



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- Cis is first cisternae.
- Trans is final cisternae.
- From trans, proteins are packaged into vesicles destined to lysosomes, secretory vesicles, or the cell surface.
- Transport vesicles are used to move back and forth between the ER and Golgi bodies
- One side of the Golgi body receives transport vesicles produced by the ER. This side is called cis face.
- Opposite side releasing vesicle is trans face.
- The Golgi body marks and sorts the molecules into different groups.
- These groups are sent in secretory vesicles inside or outside of the cell.
- Proteins are modified before sending by adding specific sugar molecules are added to a core oligosaccharide that is attached to a protein.

Water pollution

Water pollution happens when chemicals or dangerous foreign substances are introduced to water, including chemicals, sewage, pesticides and fertilizers from agricultural runoff, or metals like lead or mercury

Categories

- 1) Sewage
- 2) Disease-causing agents
- 3) Sediment pollution
- 4) Inorganic plant and algal nutrients
- 5) Organic compounds



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- 6) Inorganic chemicals
- 7) Radioactive substances
- 8) Thermal pollution

Causes water pollution: The main water pollutants include bacteria, viruses, parasites, fertilizers, pesticides, pharmaceutical products, nitrates, phosphates, plastics, fecal waste and even radioactive substances. These substances do not always change the colour of the water, meaning that they are often invisible pollutants.

However, water pollution causes diarrhea, skin diseases, malnutrition, and even cancer and other diseases related to water pollution.

Evolution of Third atmosphere of the earth

- Constant rearrangement of continental plates
- Inflow and outflow of CO₂ from large continental carbonate stores
- O₂ synthesized in photosynthesis was being used in reducing materials
- Gradually, it was shifted to oxidizing atmosphere instead of reducing
- 541 million years ago, O₂ made stable 15% of atmosphere

Type of pseudopodia

Majorly amoeba has four types

1. Lobopodia
2. Filopodia
3. Reticulopodia



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4. Axopodia

Sources of wastewater

- Domestic wastewater from households
- Municipal wastewater from communities (also called sewage)
- Industrial wastewater from industrial activities
- Agricultural wastewater from agricultural activities
- Storm water
- Surface runoff
- Sewer inflow or sewer infiltration

What is panting?

Panting, a method of cooling, used by many mammals, most birds, and some reptiles, accomplished by means of the evaporation of water from internal body surfaces.

Detail of Acid rain

Acid rain is a type of rain that contains harmful chemicals. It forms when pollutants, such as sulfur dioxide and nitrogen oxides, are released into the air from activities like burning fossil fuels and industrial processes.

When these pollutants mix with water vapor in the air, they create acids like sulfuric acid and nitric acid. When it rains, these acids come down with the raindrops, making the rain acidic. Acid rain can also fall as snow, sleet, or fog.

Affects:

Acid rain can have negative effects on the environment and our surroundings. It can harm plants, trees, and crops by damaging their leaves, stems, and roots. It can also make bodies of water, like



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lakes and rivers, more acidic, which can harm aquatic life such as fish and amphibians.

Additionally, acid rain can corrode buildings, statues, and other structures made of stone or metal. It can damage the paint on cars and buildings as well.

Prevention:

Efforts have been made to reduce the amount of pollutants released into the air to prevent acid rain.

These include using cleaner fuels, implementing stricter regulations on industries, and promoting energy conservation and alternative energy sources.

Difference between methanogen and halophiles

1. Methanogens

- They release methane (CH_4) as a waste product
- Many live in mud at the bottom of lakes and swamps because it lacks oxygen
- Some live in the intestinal tracts of animals to help break down food.
- They could play a role in garbage/sewage cleanup by eating garbage.

2. Halophiles, Halo = salt, phil = loving

- Can live in water with salt concentrations exceeding **15%**.
- The ocean's concentration is roughly **4%**.
- One habitat of halophiles is Great Salt Lake. Actually three to five times saltier than the ocean.
- **2.5** million tons of sodium chloride is extracted from the lake.



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What is omnivore and types?

Primary food source is both plant and animal-based.

Examples: human, bear, cockroach, chicken

- Digest carbohydrates, protein, fat, and fiber, and metabolize the nutrients

Types

- **Frugivores:** wolves and orangutans
- **Insectivores:** swallows and pink fairy armadillos
- **Granivores:** large ground finches and mice

How Digestion Occur in Class Pisces??

Digestion

- Herbivore, carnivore and omnivore
- Complete gut
- With stomach/without stomach
- Liver, pancreas, pyloric caeca and endocrine system help in digestion

Range of stratospheric in poles

The second major layer of Earth's atmosphere, Above the troposphere, and below the mesosphere

Height

- 20 km (66,000 ft; 12 mi) at equator
- 10 km (33,000 ft; 6.2 mi) at midlatitudes
- 7 km (23,000 ft; 4.3 mi) at the poles



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Uses of peroxisomes

- In yeast and plant cells, this process is carried out exclusively in peroxisomes.
- B-Oxidation of fatty acid
- Glyoxylate cycle
- Photorespiration (Glycolate pathway)
- Degradation of purines
- Decomposition of hydrogen peroxide

Characteristics of Phylum Mastigophora

- Unicellular or colonial
- Locomotion by flagella, pseudopodia, or both
- Autotrophic (self-nourishing), saprozoic (living in decaying organic matter), or heterotrophic (obtains energy from organic compounds)
- Single type of nucleus
- Sexual reproduction (usually)

Note on Troposphere

- The lowest layer
- All weather conditions take place here. Conventional aviation takes place here.

It contains:

- 75% of the atmosphere's mass
- 99% of the total mass of water vapor and aerosols

Height:

- 18 km (11 mi; 59,000 ft) in the tropics
- 17 km (11 mi; 56,000 ft) in the middle latitudes
- 6 km (3.7 mi; 20,000 ft) in the polar regions in winter.



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- The total average height of the **troposphere is 13 km.**

Properties:

- Composition of air is homogenous except water vapors.
- Temperature of the troposphere decreases with altitude.
- Ground absorbs most of the sun's energy
- It makes the lower levels of the atmosphere warm.
- Radiating of heat away from earth at the top of the atmosphere
- It results in the cooling of that part of the atmosphere.
- Vapor pressure drops with decrease in temperature.
- Drop in vapor pressure is due to gravity.
- Pressure of the atmosphere is maximum at sea level and decreases with altitude.

Invasive species

Invasive species are plants, animals, or other organisms that are not native to a particular ecosystem but are introduced to it, often by human activities.

Types of invasive species

1. Animals
2. Plants
3. Pathogens

These species can spread quickly and have negative impacts on the environment, economy, and even human health.

Invasive species can be introduced to a new environment through various means, such as:

- accidental or intentional release
- escape from captivity
- Hitchhikers on ships, planes, or vehicles.



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- Seed dispersal by birds
- Lowering of sea levels join two isolated land masses
- Flooding

Once they are introduced, they can spread quickly and easily because they have no natural predators or diseases to control them. They can also adapt to new environments and outcompete native species for resources such as food, water, and space.

Cause:

Invasive species can cause harm to the environment by altering habitats and changing the way ecosystems work.

For example, invasive insects can damage crops, invasive plants can reduce timber production, and invasive animals can reduce fish populations.

These impacts can cause financial losses for industries such as agriculture, forestry, and fisheries

Prevention:

Preventing the introduction and spread of invasive species is important to protect native species and ecosystems. This can be done through measures such as monitoring and early detection, controlling invasive species populations, and educating the public on the risks of introducing non-native species

Gene flow

- Gene flow can be either a constructive or a constraining force.
- It can spread a beneficial mutation that arises in one population to other populations.
- Gene flow can impede adaptation within a population by the continual flow of inferior alleles from other populations.



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- Natural selection is removing inferior alleles while gene flow can bring them back.

Different methods of water pollutants

Persistent Water pollutants are:

- Some pesticides (DDT, dieldrin)
- Some leachate components from landfill sites (municipal, industrial)
- Petroleum and petroleum products (Oil pollution)
- Heavy metals such as lead, mercury, cadmium
- Radioactive materials such as strontium-90, cesium-137, radium-226, and uranium

Characteristics of Phylum Echinodermata

- 7000 species are exclusively marine
- Penta radial symmetry
- Skeleton consists of a series of calcium carbonate plates called **Ossicles**
- Water vascular system
- Series of water-filled canals, and their extensions called **tube feet**
- Hemal system derived from coelomic cavities
- Nervous system consist of nerve net, nerve ring and radial nerves
- Complete digestive tract
- Ammonia is the primary nitrogenous waste product
- Regeneration is common.
- Regeneration of lost arms
- Reproduction is Dioecious



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- External fertilization

Note on carbon cycle

The carbon cycle is the process by which carbon moves from the atmosphere into living organisms and the Earth, and then back into the atmosphere again.

Carbon is main component of living organisms. **20%** of human body is carbon by weight. CO₂ is major source of carbon. Atmosphere contains about **750 billion** metric tons of carbon in form of CO₂.

Photosynthetic organisms take in CO₂ from the atmosphere

Synthesize the carbon containing organic compounds This process is sometimes called carbon fixation. Animals eat plants and make their own organic compounds. Production of energy by breakdown of these organic compounds in respiration.

Respiration releases CO₂ in air. Decaying organisms also produce CO₂.

CO₂ released in air is again available for plants. In aquatic ecosystem, CO₂ reacts with the water to form bicarbonate ions. Dissolved CO₂ and bicarbonates are used by algae and aquatic plants in photosynthesis.

Evolutionary changes? an example of cytochrome C

- Evolutionary changes whether over long period or rapid are result of change in genetic makeup.
 - DNA
 - Changes in the base sequences in DNA and amino acids in proteins



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Cytochrome c is a protein present in the cellular respiration pathways in all eukaryotic organisms.

- Closely related organisms have similar cytochrome c molecules.
- Structure of cytochrome c has changed so little during hundreds of millions of years.
- It is said to have been conserved evolutionarily. Highly conserved proteins can help establish evolutionary relationships among distantly related organisms.
- Not all proteins are conserved as rigorously as **cytochrome c**.

How echinoderms regenerate?

Regeneration in Echinodermata:

All echinoderms have the capability to regenerate some or most organs or appendages. Echinoderms, such as starfish, have the incredible ability to regenerate their entire body from just a portion of the central disk.

If a starfish's arm containing a portion of the central disk is broken off, it can regenerate a whole new organism from that fragment. This process of regeneration is facilitated by the starfish's unique cellular properties and genetic mechanisms.

Note on R- selected species

R-selected species:

- High growth rates
- Typically exploit less crowded ecological habitats
- Produce many offspring
- Relatively low probability of surviving to adulthood
- High reproductive rate supports their survival.



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- Especially in unstable environment

Traits of r-selected species

- High rate of reproduction
- Small body size
- Early maturity onset
- Short generation time
- Ability to disperse offspring widely

Examples: bacteria, grasses, cephalopods, rodents

K-selected species

- Living at densities close to carrying capacity
- Typically are strong competitors in such crowded habitats.
- Invest more heavily in fewer offspring
- Relatively high probability of surviving to adulthood
- Survive in stable environment
- Ability to compete successfully for limited resources is crucial
- Populations of k-selected organisms typically are very constant in number.
- Close to the maximum that the environment can bear

Traits of k-selected species

- Large body size
- Long life expectancy
- Production of fewer offspring
- Often require extensive parental care until they mature

Examples: humans, whales, parrots, eagles

Explain the Characteristics of Nematodes?

- Triploblastic
- Bilateral



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- Vermiform
- Unsegmented
- Pseudo coelomate
- Complete digestive tract
- Unique excretory system comprised of two or more renette cell
- Body wall has only longitudinal muscles
- Cuticle may contain spines, bristle papillae and ridges
- Pseudocoelom is a spacious fluid filled cavity
- Visceral organs
- Form hydrostatic skeleton
- May be herbivores, carnivores, omnivores and saprotrophs
- Feed on blood and tissue fluid of their hosts
- Complete digestive tract
- Excretion of nitrogenous waste
- Ventral gland cells, called renette
- Nervous system consist of anterior nerve ring

Reproduction

- Dioecious and dimorphic
- Males are smaller than females

Examples

- Ascaris
- Roundworms
- Hookworms
- Heartworms

Frequency Dependent Selection

This type of selection favors certain phenotypes depending on how commonly or uncommonly they occur.



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It has two types:

1. Negative
2. Positive

Negative frequency dependent selection:

- Rare phenotypes are favored by selection
- Making them more common thus maintaining variation.

Example:

- Fish predation on an insect, the water boatman, which occurs in three different colors. Experiments indicate that each of the color types is preyed upon disproportionately. Fish eat more of the common-colored insects than uncommon colors.

Positive frequency-dependent selection:

- Favors common forms
- It tends to eliminate variation from a population.

Example:

- Predators don't always select common individuals.

Types of Consumers

1. Herbivore
2. Carnivore
3. Omnivore

1. Herbivore

- Primary food source is plant-based.

Examples: vertebrates as well as invertebrates



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Digestive systems capable of handling large amounts of plant material. There are **eight** types of herbivory depending upon different feeding habits.

2. Carnivore

- Primary food source is animal-based.

Examples: vertebrates as well as invertebrates

- Digestive systems are more efficient in protein digestion.

2. Omnivore

- Primary food source is both plant and animal-based.

Examples: human, bear, cockroach, chicken

- Digest carbohydrates, protein, fat, and fiber, and metabolize the nutrients

Types of Omnivores:

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What is nucleus made up of?

1. The nuclear envelope
2. Nucleoplasm and Chromatin (two major types of material)
3. The nucleolus
4. DNA

Two meters of human DNA fits into a nucleus that's 0.000005 meters across Metamorphosis and its types



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Exponential Growth & Logistic Growth

Exponential growth	Logistic growth
It occurs when resources are abundant	It occurs when resources are limited
Population passes well beyond the carrying capacity of the ecosystem	Population doesn't grow beyond the carrying capacity of the ecosystem
It produce J shaped curve	It produce S shaped curve
A stationary or stable phase is completely achieved	A stationary or stable phase is achieved
It has two phases lag and log	It has four phases lag, log, deceleration and steady
It occurs in fewer organisms like lemmings etc.	It is more common in wildlife, yeast etc.
$DN/dt = Nr$ N= population size, r= intrinsic rate	$DN/dt = rN(k-N)/K$ DN/dt= change in population size per unit time K= carrying capacity

Pollution

Introduction of contaminants into the natural environment that cause adverse change. Undesirable change in physical, chemical, or biological characteristics of any component of the environment.

Pollution Classified as:

1. Point source
2. Non point source

1. Point source

Single identifiable source of pollution

Example:

- Land fill
- Factory drain
- Sewages pipes



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2-Non Point source

Resulting from many diffuse sources

Example:

- Sediment
- Nutrients
- Toxic contaminants and chemicals
- Pathogens

Types of Pollution

- Land
- Air
- Water
- Light
- Noise

Detail notes on Type of pseudopodia

Majorly amoeba has four types

1-Lobopodia

- Contain ectoplasm and endoplasm
- Used for locomotion
- Used for engulfing feed

2-Filopodia

- Only ectoplasm in pseudopodia
- Two-way streaming
- Provides food in conveyor belt fashion



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3-Reticulopodia

- Similar to filopodia
- Branch and rejoin to form net like series of cell extensions

4-Axopodia

- These are thin filamentous supported by central axis of microtubules.
- Food caught on axopodia delivered to the central cytoplasm of amoeba.

Differentiate between monogamous, polygamous and polyandrous?

Monogamous:

A mating system where an individual has only one partner at a time.

Polygamous:

A mating system where an individual has multiple partners of the opposite sex at the same time.

Polyandrous:

A mating system where a female has multiple male partners at the same time.

Mosaic Evolution:

A species is a mosaic of different molecules and structures that have evolved at different rates. Some molecules or structures are conserved in evolution.



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Others change more rapidly. The basic design of a bird provides a simple example. All birds are easily recognizable as they because of highly conserved structures:

1. Feathers
2. Bills
3. Body form

Difference between grazer and browsers

Different feeding strategies:

- Grazers
- Browsers

1. Grazers

Feeds on plants such as grasses, soft flowering plants (forbs) and algae. Plants are not killed.

Examples: Deer & Snails

2. Browsers

Feed on non-grass plant material

Leaves, soft shoots, twigs, fruits, woody plants such as shrubs

Examples: Goats, deer, giraffe, elephants

Types of Ecological Pyramids

An ecological pyramid is a graphical representation of the relationship between different organisms in an ecosystem. It shows the number, biomass, and energy of different trophic levels in an ecosystem.



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Types:

- 1) **Pyramid of energy:** represents how much energy is available in each trophic level
- 2) **Pyramid of numbers:** represents the actual number of organisms present in each trophic level
- 3) **Pyramid of biomass:** represents the total mass of living things in each trophic level

Write The Origin of Eukaryotic Cells

According to Endosymbiotic Theory:

- Chloroplasts and Mitochondria evolved from symbiotic bacteria that lived inside of a larger prokaryote.
- Also known as Symbiogenesis
- Chloroplasts evolved from endocytosis of cyanobacteria
- Mitochondria evolved from endocytosis of aerobic bacteria.

According to Evidence of evolution of mitochondria:

- Membranes- Have double cell membranes
- DNA- Have circular DNA, like bacterial genome
- Reproduction- via budding, just like bacteria
- Size- similar size to bacteria (1-10 microns)
- Multicellularity
- Cell specialization, larger size, more complexity
- Volvox algae with division of labor

Primary and secondary treatment of wastewater?

Primary Treatment:

- Mechanical process
- Screening
- Grinding



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- Grit Chamber (heavy particles - “grit”)
- Sedimentation Tank (suspended solids - “sludge”)
- Chlorination of effluent

Secondary Treatment:

- Biological process
- Uses bacteria to remove organic matter dissolved in water (ODW) and lower BOD

Types

- Trickling Filter
- Activated Sludge

Characteristics and classification of phylum Platyhelminthes

Characteristics:

- Usually flattened dorsoventrally
- Triploblastic
- Acoelomate
- Bilaterally symmetrical
- Unsegmented worms
- Beginning of cephalization with an anterior cerebral ganglion
- Longitudinal nerve cords
- Protonephridia as excretory structures
- Hermaphroditic
- Complex reproductive systems
- Reproduce asexually by transverse fission
- Monoecious
- Cross-fertilization



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Classification

- 1) Class Turbellaria (Free-Living Flatworms)
- 2) Class Monogenea
- 3) Class Trematoda (Flukes)
- 4) Class Cestoidea (Tapeworms)

What is Aestivation?

Survival in extended period of drying. Avoid temperature damage and dehydration

Examples: Insects, frog, turtle

Different layers of Atmospheres

Temperature shows different patterns. It increases with altitude in some regions. Temperature is useful tool to distinguish atmospheric layers. Atmosphere can be divided into five layers. Lowest to Highest Layers of atmosphere:

1. Troposphere: 0 to 12 km (0 to 7 miles)
2. Stratosphere: 12 to 50 km (7 to 31 miles)
3. Mesosphere: 50 to 80 km (31 to 50 miles)
4. Thermosphere: 80 to 700 km (50 to 440 miles)[12]
5. Exosphere: 700 to 10,000 km (440 to 6,200 miles)

Effects of Global warming

- Physical Environmental Effects
- Effects on Biosphere
- Effects on Humans
- Arctic sea ice shrinking
- Global sea level rise
- Non-polar glaciers
- Continental Precipitation



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- Extreme Weather conditions
- Changes in ocean properties
- Changes in Ecosystem
- Impacts on Humans
- Recent Climatic Trends in Pakistan (Department of Meteorology)

Mitochondria

- Mito – thread
- Chondrion – granule like
- First observed by Richard Altman (1894)
- Term mitochondria was coined by Carl Benda (1898).

Size: 0.05 – 1.0 μm

Length: 1 – 10 μm long

Shape: Bean shaped, in fibroblast it is elongated and thread like.

Number: Depends on type, size and functional state of cell.

Example: average liver cell contain around **1500** mitochondria.

Structure:

- Outer Membrane
- Intermembrane Spac (Perimitochondrial space)
- Inner Membrane
- Cristae
- Matrix

What is cloud pollution?

Cloud pollution, also known as cloud computing pollution, refers to the environmental impact caused by the energy consumption and



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carbon emissions associated with data centers and cloud computing infrastructure

Adaptedness or fitness:

Is a measure of the capacity for successful reproduction in a given environment.

- Not every characteristic is an adaptation to some kind of environmental situation.
- Neither adaptations lead to perfection.
- Fitness is quantified as reproductive success
- The number of surviving offspring left in the next generation.
- The most fit phenotype is simply the one that produces, on average, the greatest number of offspring.

Ecological niche?

Each organism in a community confronts the challenge of survival in a different way. Occupies a specific functional role and place called its niche. The niche an organism occupies is the total of all the ways it uses the resources of its environment.

A niche may be described in terms of:

- Space utilization
- Food consumption
- Temperature range
- Appropriate conditions for mating
- Requirements for moisture

What Is the Role of University of Wisconsin Madison Arboretum in Restoration ecology?

- The restoration of the prairie was at an early stage in November 1935.



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- The prairie as it looks today. This picture was taken at approximately the same location as the 1935 photograph

How evolution and paleontology is related to each other?

Paleontology:

Paleontology is the study of fossils, which are the preserved remains or traces of ancient organisms. Paleontologists use fossils to reconstruct the history of life on Earth and to understand how organisms have changed over time.

Paleontologists use a variety of techniques to study fossils, including carbon dating, radiometric dating, and comparative anatomy.

Evolution:

Evolution is the process by which species change over time through the process of natural selection. It is supported by a wide range of evidence from many different fields of biology, including genetics, ecology, and comparative anatomy.

Differentiate between Mimicry and Camouflage?

Mimicry:

- The resemblance of one species to another for protective or aggressive purposes
- Different from camouflage
- They imitate instead of hiding by camouflage strategies
- **Example:** poisonous chemicals in insects e.g. Butterflies



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Camouflage:

- Use of any combination of materials, coloration, or illumination
- Concealment from predators
- Also known as cryptic coloration
- **Example:** Egyptian nightjar, Parrot

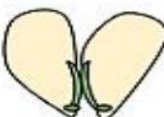
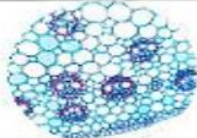
Different types of flagella?

- Motility - movement

Arrangement basis for classification

1. Monotrichous; one flagella
2. Lophotrichous; tuft at one end
3. Amphitrichous; both ends
4. Peritrichous; all around bacteria.

Difference b/t Monocot & Dicot plants

MONOCOT		DICOT	
Single Cotyledon		Two Cotyledon	
Long Narrow Leaf Parallel Veins		Broad Leaf Network of Veins	
Vascular Bundles Scattered		Vascular Bundles in a Ring	
Floral Parts in Multiples of 3		Floral Parts in Multiples of 4 or 5	



Zoology Legends

Reproductive system of arthropods

In terms of reproduction, most arthropods are dioecious and have paired genital openings. Males often package sperm in a spermatophore and transfer them into the female genital tract during copulation. Sperms are transferred via modified pedipalps.

Define Microfilament and it's function

Microfilaments: (actin filaments)

Microfilaments, also known as actin filaments, are thin structures composed of actin proteins that form a 3-D network inside the cell membrane.

Structure:

- Thinnest class of fibers
- Solid rods of protein, actin
- Twisted double chain of actin subunits
- About 7 nm in diameter

Functions:

- 3-D network inside cell membrane
- In muscle cells, actin filaments interact with myosin filaments to create muscle contraction.
- Cell movement
- Intracellular transport/trafficking
- Maintenance of eukaryotic cell shape
- Cytokinesis
- Cytoplasmic streaming



Zoology Legends

Digestion of echinoderms

Digestive System

- Complete digestive tract that may be secondarily reduce
- Mouth open to a short esophagus
- Large stomach
- Stomach is divided into two region
- Larger, oral stomach (cardiac stomach)
- Smaller, aboral stomach (pyloric stomach)
- Absorptive structure called pyloric cecae

What is mutualism?

Both species coevolve to benefit each other in same habitat

Example: Wasp Relationship

Evolution of Mesosphere?

- Layer, directly above the stratosphere and directly below the thermosphere
- Lower boundary is 50 to 65 kilometres
- Upper boundary is around 85 to 100 kilometres
- It is called as “near space”.
- Temperature decreases with increasing height
- Due to decreasing absorption of solar radiation

Which diseases are caused by apicomplexan?

Here are some diseases caused by apicomplexan parasites

- 1) Malaria - caused by the parasite Plasmodium.
- 2) Toxoplasmosis - caused by the parasite Toxoplasma gondii.



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- 3) Babesiosis - caused by various species of the parasite Babesia.
- 4) Cryptosporidiosis - caused by the parasite Cryptosporidium.
- 5) Coccidiosis - caused by the parasite Eimeria

Circulatory system of nematodes?

Nematodes have a relatively simple circulatory system. They lack a true circulatory system with specialized blood vessels, but instead rely on a fluid-filled body cavity called the pseudocoelom to transport nutrients and gases. This fluid, called pseudocoelomic fluid, helps distribute nutrients and oxygen throughout the body and aids in waste removal.

However, nematodes do not have a dedicated organ for pumping and circulating this fluid like a heart.

Which limiting factor effects population size?

Factors limit the population size and growth.

1. Density dependent factors
2. Density independent factors

Density dependent factors

Example: Food Competition within the population.

When a population reaches a high density, there are more individuals trying to use the same quantity of resources. This can lead to competition for food, water, shelter, mates, light, and other resources needed for survival and reproduction.

Predation

Higher-density populations may attract predators who wouldn't bother with a sparser population. When these predators eat



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individuals from the population, they decrease its numbers but may increase their own.

Disease and Parasites

Disease is more likely to break out and result in deaths when more individuals are living together in the same place.

Waste Accumulation

High population densities can lead to the accumulation of harmful waste products that kill individuals or impair reproduction, reducing the population's growth.

Density Independent Limiting Factors

Affect per capita growth rate independent of density of population

- Natural disasters
- Severe weather
- Pollution Cannot regulate population at constant level . They are accidental. Cause abrupt shift in population size. May wipe out small population by sporadic event
- Collectively, density dependant and independent factors create a different population growth curve.
- Cyclic oscillations

Type of Zoomastigophore

- Lack chloroplast
- Heterotrophic or saprozoic
- Parasites of humans



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Examples:

1. *Trypanosoma brucei*

- It has three sub species.

1) *T. b. brucei*:

- Parasite of non-human
- Common host are mammals of Africa

2) *T. b. gambiense*:

- Causes sleeping sickness in humans

3) *T. b. rhodesiense*:

- Causes sleeping sickness

Characteristics of chordates

- Chordates have a notochord, a flexible rod-like structure that provides support.
- They possess a dorsal nerve cord, which develops into the spinal cord in vertebrates.
- Chordates have pharyngeal slits or clefts, which may function as respiratory structures or filterfeeding mechanisms.
- They have a post-anal tail that extends beyond the anus.
- Chordates have an endostyle or thyroid gland, which plays a role in metabolism and hormone regulation.
- Chordates are a diverse group of animals.
- They have a bilateral symmetry.
- Chordates have a coelom, a body cavity that contains their internal organs.
- They possess a closed circulatory system.
- Chordates have a well-developed nervous system.
- Chordates exhibit segmentation, with their bodies divided into repeating units.



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- They have a complete digestive system with a mouth and an anus.
- Chordates possess a well-developed muscular system for movement.
- They have a high level of cephalization, with a concentration of sensory organs and a distinct head region.
- Chordates exhibit a wide range of reproductive strategies, including internal fertilization and external egg-laying.

Reproduction in Arthropods

Arthropods reproduce by sexual reproduction, which involves the generation and fusion of gametes. Most arthropods are either male or female, and they undergo internal fertilization. Once the egg has been fertilized, the female usually lays the egg, and it continues developing outside of the mother's body.

However, some arthropods are hermaphroditic, meaning they have the organs of both sexes, such as in barnacles. Few crustaceans and insects can reproduce by means of parthenogenesis. They revert to reproducing sexually when conditions become more favorable again.

Theory of inheritance and soft inheritance

The theory of "inheritance of acquired characteristics" or "soft inheritance" is the idea that traits that are acquired by an organism during its lifetime can be passed on to its offspring. This theory was proposed by Jean-Baptiste Lamarck in the early 19th century.

However, this theory has been largely discredited by modern genetics and evolutionary theory, which suggest that genetic



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information is passed down through DNA and that acquired characteristics are not inherited.

- If an organism developed a characteristic feature through adapting to a new way of life during its lifetime, it would pass this on to its offspring
- The classic example given is that of the giraffe's neck. As the giraffe's ancestors searched for a richer food supply they stretched to reach higher branches in trees. Thus their stretched bodies were passed onto their offspring.

Explain Methods of wastewater treatment?

It is the process of converting wastewater – water that is no longer needed or is no longer suitable for use – into bilge water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

Wastewater treatment is the process of removing contaminants and pollutants from used water to make it safe for discharge back into the environment or for reuse. It involves a combination of physical, biological, and chemical processes to remove solids, organic matter, nutrients, and pathogens, ensuring the protection of public health and the environment.

Processes:

- 1) Primary
- 2) Secondary
- 3) Advanced



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Primary treatment:

Primary treatment in wastewater treatment involves mechanical processes such as screening, grinding, grit chamber, and sedimentation tank. Screening and grinding remove large debris and break down solids, while the grit chamber settles heavy particles and sedimentation tanks separate suspended solids.

Chlorination may be used to disinfect the effluent. Primary treatment reduces pollutant load but is typically followed by secondary and tertiary treatment for further purification.

Secondary treatment:

In secondary treatment using trickling filters, the wastewater from the sedimentation tank is passed through a bed of stones, where bacteria grow and consume organic matter. Cleaner water then trickles through pipes at the bottom for further treatment.

In secondary treatment using activated sludge, the wastewater is pumped into an aeration tank and mixed with air and bacteria-rich sludge called "activated sludge." The bacteria decompose the raw sewage, reducing organic and biological oxygen demand. The mixture, called mixed liquor, then goes to another sedimentation tank where suspended solids settle out for reuse as activated sludge.

These processes help further purify the wastewater before its safe release or reuse.

Advanced tertiary treatment:

Advanced tertiary treatment involves using physical and chemical processes to target specific pollutants that remain in wastewater after primary and secondary treatment. This treatment is quite expensive, costing twice as much as building a secondary treatment



Zoology Legends

plant. The processes used in advanced tertiary treatment include bleaching to remove coloration, disinfection to kill pathogens, coagulation-sedimentation with alum, adsorption using activated charcoal, and electrodialysis for salt removal. These processes help ensure that the wastewater is thoroughly treated before it is released back into the environment.

Write a note on binomial nomenclature?

- The binomial system of nomenclature brings order to a chaotic world of common names.
- Common names vary from country to country, and from region to region within a country.
- There was need of globally recognized names.
- In binomial system, no two kinds of animals have the same binomial name.
- Every scientific name has two parts:
 - 1) First from genus
 - 2) Second from species
- Names are written in italics

Example:

- Common name: Indian frog
 - Genus name: *Rana*
 - Species name: *tigrina*
 - Scientific name: *Rana tigrina*
- This scientific name is universally accepted.

Q. Define parasitism?

Parasitism is a type of symbiotic relationship between two species in which one species, the parasite, benefits at the expense of the other species, the host



Zoology Legends

Q. Explain polymorphism with example?

- Polymorphism occurs in a population when two or more distinct forms exist Without a range of phenotypes between them. Occurrence of distinct variations within a population, without intermediate or overlapping phenotypes. It highlights the genetic diversity and variability within a species.

Balanced Polymorphism

It occurs when different phenotypes are maintained at relatively stable frequencies in the population. Resemble a population in which disruptive selection operates.

Example:

- Sickle cell anemia is homozygous recessive trait.
- Most common in Africa where malaria is common as well.
- Heterozygous alleles carriers are less more susceptible to malaria.
- Homozygous dominant alleles carriers are more susceptible to malaria.
- Homozygous recessive alleles carriers have low survival due to anemia.
- Therefore, heterozygous are more successful in survival and reproductive potential than two others.

Q. Write three classes of phylum reptiles?

- 1) **Order Testudines:** Tortoises, turtles
- 2) **Order Squamata:** Snakes, lizards, worm lizards, Lizard like reptiles, single species, Sphenodon and Tuataras
- 3) **Order Crocodilia:** Crocodiles, alligators, gavials



Zoology Legends

Q. Explain types of parasites?

Q. Explain classification and characteristics of phylum platyhelminthes?

Ozone depleting substance

Peroxisome

Genetic drift types

Long

Analogy homology

Feeding level of food chain

Binomial nomenclature

Endomembrane system

3MARKS

- ☐ justify how the 1 of fourteen different species on Darwin finches on the Galapagos Island related to the allopatric speciation?
- ☐ Crypsis define with the change of skin color and counter shading
- ☐ Name the factors control of an ecosystem
- ☐ Why theory of inheritance of acquired characteristics or soft inheritance rejected?
- ☐ Functions of Peroxisomes?
- ☐ Note on Troposphere
- ☐ Note on Water Pollution
- ☐ Digestive System of Echinodermata



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- ☐ Explain rate of Evolution



ZOOLOGY
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Muhammad Zaman

03044756496

Remember me in you Precious Prayer

Jazak Allah